When Do Native Speech-Language Models Beat Cascades?

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Motivation

- Conversational AI has **transitioned from text-only models to native, end-to-end speech-to-speech (S2S) frameworks**
- Modern voice agents integrate acoustic and linguistic features directly, bringing human-like emotional expressiveness (EQ) and natural dialog flows to users
- Commercial and open-source models have **achieved massive scale** (e.g., Doubao surpassing 200 million monthly active users),transforming telecommunications, customer support, and direct human-machine collaboration.

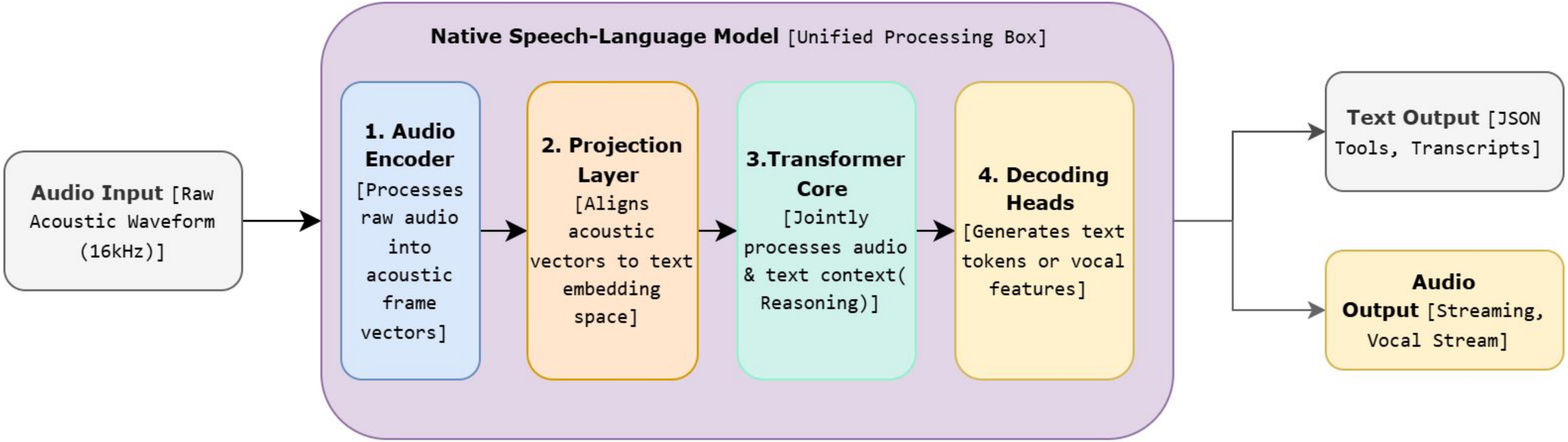
State-Of-The-Art Voice Agents

ChatGPT-4o
- Native Voice-Voice
- Full-Duplex Chat
- Native Tool-Use

Doubao
- Over 200M Users
- High-EQ Expressiveness
- Ultra-Low Latency

Qwen-Omni
- Open-Source SOTA
- “Think-Talker” MoE
- Dynamic Reasoning

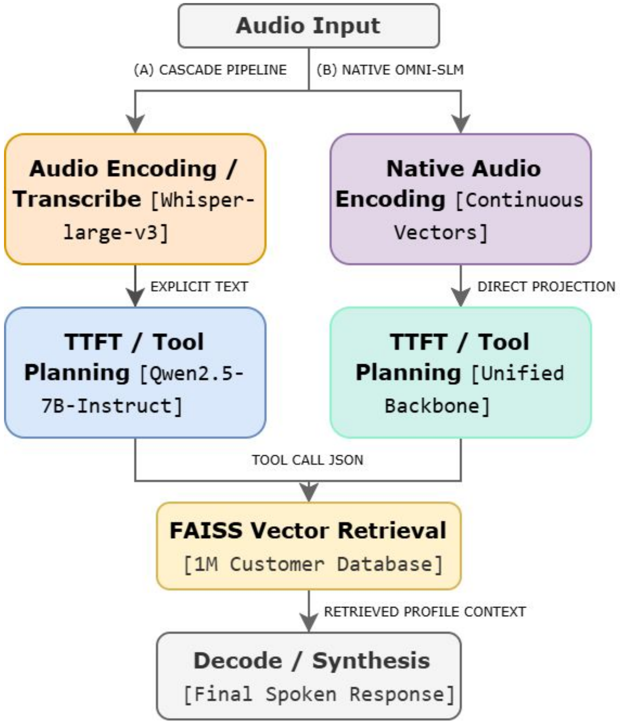
How Native Speech-Language Models Work



- Raw wave signals bypass text transcriptions (ASR) entirely, mapping acoustic frames straight into the model's linguistic embedding space.

Problem

- Traditional speech assistants rely on cascaded pipelines (ASR → LLM)
- These modular architectures suffer from severe error propagation (A single typo in the speech step ruins the reasoning step)
- Chaining multiple models causes high latency
- Direct speech models promise to fix this but their limits are untested



Production Requirements

To work in the real world, a voice assistant must master 3 areas:

- Tone and Meaning
  - Understanding both spoken words and voice emotions/nuances.
- Perfect Accuracy
  - Parsing phone numbers and IDs without a single typo.
- Fast Responses
  - Keeping delays low enough for a natural, real-time conversation.

Research Question

Can native Omni-SLMs **replace** traditional cascaded pipelines **under these stringent enterprise constraints**?

Experiment Setup

Models Under Test

1. **Cascade Baseline** (ASR +LLM)
   a. Whisper-large-v3
   b. Qwen2.5-7B-Instruct
   c. Served via vLLM token
2. **Qwen3-Omni-30B-AWQ** (Native Omni-SLM)
   a. Alibaba
   b. 31.6B total parameters
   c. 4-bit AWQ quantized
3. **Nemotron-3-Nano-Omni-30B-BF16**
   a. NVIDIA
   b. 31.6B Total, 3.6B active parameters
   c. MoE BF16

Dataset & Infrastructure

Table 1: Acoustic and Demographic Parameters of the $N = 500$ Evaluation Dataset

Acoustic Parameter	**Value / Configuration**
Sampling Frequency	$16.0\,\mathrm{kHz}$ (general queries, resampled); native engine rate (lookup queries)
Resolution	16-bit Linear PCM
Channel Mode	Mono
Average Utterance Duration	$4.82\,\mathrm{s}$
Spoken Format	Standard E.164 (International)
Stratification Ratio	20 / 80 (Stratified Interleaved)
TTS Engines	gTTS (general queries); pyttsx3 (lookup queries)

Accuracy & Alignment Results

Table 2: Operational Accuracy, Vector Alignment, and Hallucination Suppression Profiles ($N = 500$ audio samples per model; see note below for accuracy-scoring denominators)

Model	Tool Prec.	Tool Recall	Tool F1	TP	FP	FN	HR@1	Halluc. Score
Cascade Baseline	**0.957**	0.890	**0.922**	89	**4**	11	0.978	**0.116**
Qwen3-Omni-30B-AWQ	0.850	**1.000**	0.919	**96**	17	**0**	**1.000**	0.135
Nemotron-3-Nano-Omni-BF16	1.000	0.970	0.985	97	0	3	0.856	0.009

Note: Accuracy denominators (TP+FP+FN+TN) reflect samples that completed tool-selection scoring: 500 / 496 / 502 for Cascade / Qwen3-Omni / Nemotron, respectively. Nemotron's substantially longer per-sample runtime produced 505 raw latency traces (Table 3), of which 502 completed full accuracy scoring; raw per-sample logs were not retained to isolate the exact cause of this 3-sample gap.

Table 3: Latency Performance Breakdown and Quantiled Tail Latencies (ms)

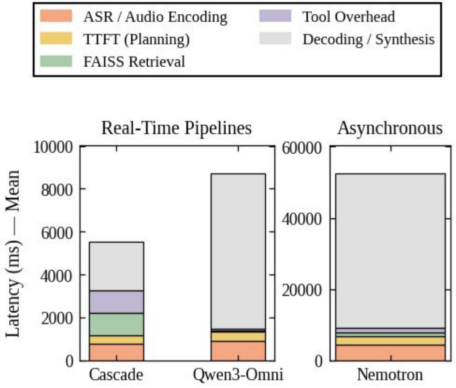
Model	Mean E2E	Median p50 E2E	p95 E2E	ASR / Enc	TTFT	FAISS	Tool Exec	Decoding
Cascade Baseline	**5,529**	**3,655**	**14,916**	**788**	**364**	1,047	1,065	**2,137**
Qwen3-Omni-30B-AWQ	8,732	6,948	20,381	915	407	**67**	**67**	4,056
Nemotron-3-Nano-Omni-BF16	52,593	39,287	132,815	4,290	2,502	1,098	1,116	31,577

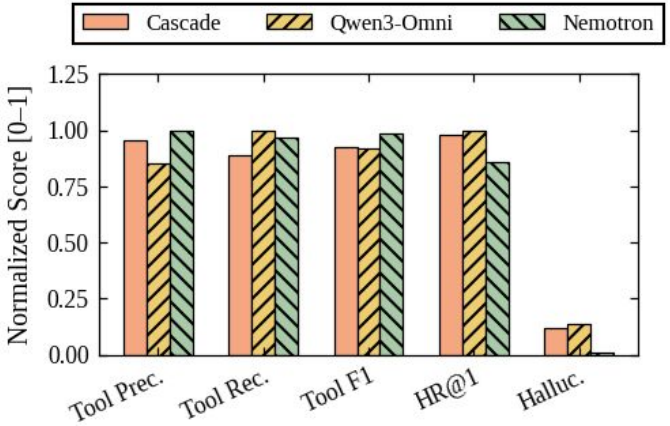
Accuracy & Alignment Results

- **Nemotron wins on tool intent,** Achieved an F1-score of 0.985 with 0 false positives.
- **Qwen3-Omni wins on retrieval,** Secured a 1.000 HR@1 database hit rate.
- **Nemotron leads fact grounding,** delivered a near-0 hallucination score (0.009)

Latency Analysis

- Near-zero hallucinations (0.009) and high tool precision (Nemotron) incur a 9.5x latency penalty (~52.6s), leaving Cascade and quantized models as viable options for real-time customer deployment (<9s)





Where should each architecture be deployed in production?

Cascade Baseline (Whisper-large-v3 + Qwen2.5-7B)

- Should be deployed in **Real-Time Voice Agent Analytics**
- Target Use Case: Live, turn-taking customer calls, where speed is the highest priority
- It has an extremely fast response times (median delay of **3.66 seconds**), fitting safely within natural human conversational limits.
- **However, ASR mistakes pass directly to the LLM with no way to recover. Requires extra text-based guardrails to prevent hallucinations.**

Qwen3-Omni-30B-AWQ (Native Speech Model)

- Should be deployed in **Real-Time Retrieval Agent**
- Target Use Case: Complex, database-reliant lookups where retrieving the correct record is critical.
- It has an flawless vector database alignment (**1.000 HR@1 hit rate**) with highly competitive serving speeds (**median delay of 6.95 seconds**).
- **However, slightly prone to over-triggering tool calls (17 False Positives) and minor quantization-driven hallucinations.**

Nemotron-3-NanoOmni-30B-BF16 (MoE)

- Should be deployed in **Offline Audit & Analytics**
- Target Use Case: Asynchronous post-call processing, call-log validation, and quality assurance auditing.
- It has a near-perfect classification logic (**0.985 F1-score**) and an exceptionally low hallucination rate (**0.009**).
- **However, extremely slow response times (median delay of 39.3 seconds) driven by unquantized Mixture-of-Experts (MoE) routing and memory bottlenecks.**

Key Findings

- **Omni-SLMs are NOT strictly superior**
- **Quantization is essential**
- **Task-specific deployment matters**
- **BERTScore underestimates digit errors**

Conclusion

For production-ready voice assistants, enterprises should adopt a hybrid, task-specific taxonomy:

- **Cascade Pipelines:** Prioritize latency for live, interactive phone routing.
- **AWQ Omni-SLMs:** Prioritize retrieval recall for real-time, data-grounded lookup agents.
- **BF16 Omni-SLMs:** Prioritize grounding accuracy for offline, high-fidelity data auditing and QA.

Future Directions

1. **Noisy Environment Testing:** Assess unified audio encoder performance under real-world distortions.
2. **End-to-End Voice-to-Voice Evaluation:** Benchmark voice-out models that synthesize raw waveforms directly, bypassing text generation.
3. **Entity-Aware Hallucination Metrics:** Combine BERTScore semantic similarity with rule-based string matching for numeric verification.

Thank You!